

CSE 350

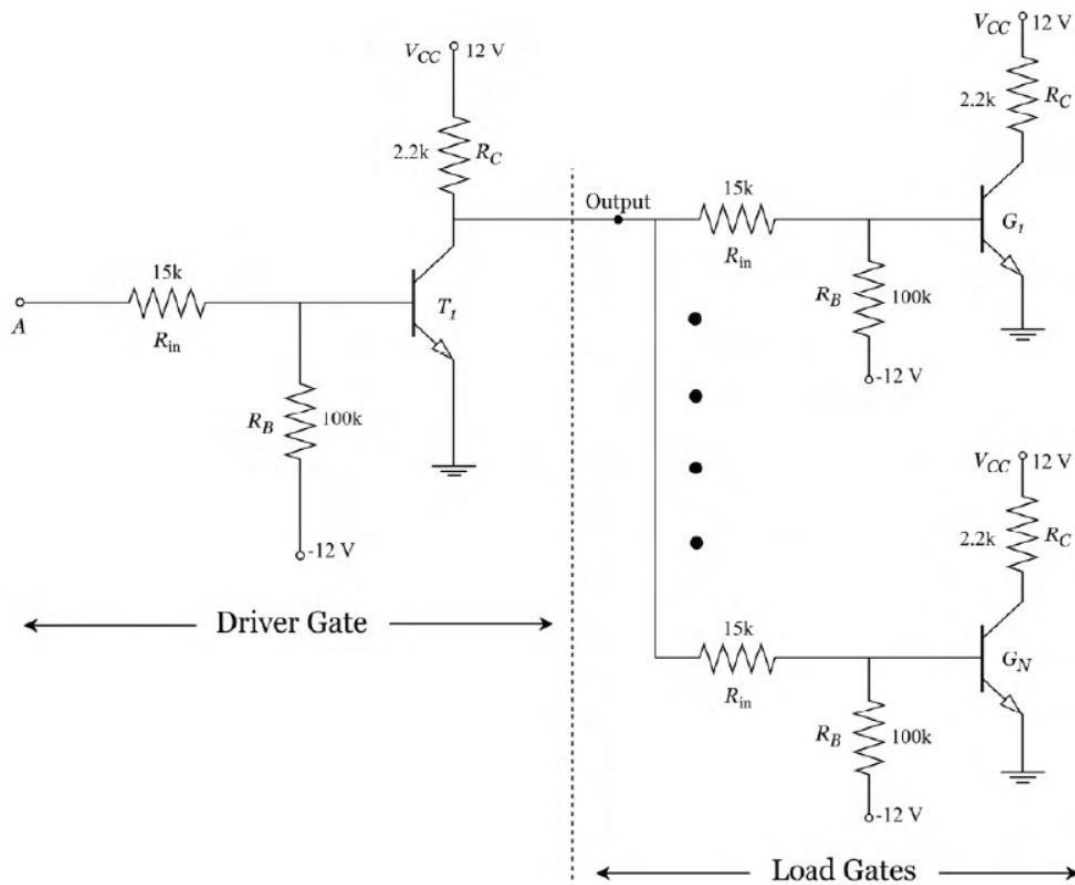
Digital Electronics and Pulse Techniques

Assignment-03

Total – 20 marks

Question 1

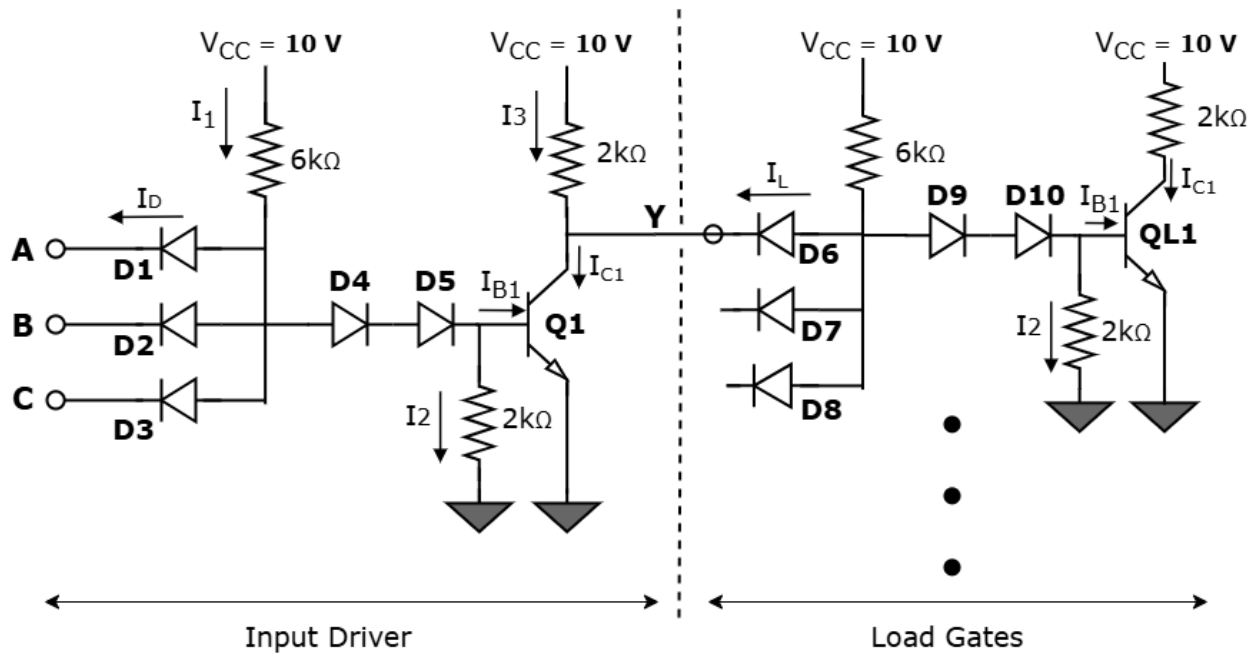
For the given RTL inverter circuit assume $V_{OH} = 10\text{ V}$ and $V_{OL} = 0.2\text{ V}$. Also assume common emitter current gain, $\beta_F = 30$. Assume for saturation mode $V_{BE} = 0.8\text{ V}$, $V_{CE} = 0.2\text{ V}$ and cut in voltage for transistor, $V_\gamma = 0.5\text{ V}$.



(a)	Find the Maximum possible Fanout.	1
(b)	Find the value of V_o if Fanout, $N=2$ (2 Load gates are connected) and input of Driver is Low.	1
(c)	If $V_{in} = \text{High}$, find the power dissipation in the Driver circuit. (assume No Loads are connected)	1
(d)	If $V_{in} = \text{High}$, find the power dissipation in the Driver circuit. (assume 50 Loads are connected)	1
(e)	Find the value Noise margin	1

Question 2

For the given circuit, $\beta_F = 30$, $V_{CE}(\text{sat}) = 0.1 \text{ V}$, $V_{BE}(\text{sat}) = 0.8 \text{ V}$



(a)	Find I_1 , I_2 , I_{B1} , I_{C1} and I_D if any input is logical low (0.1 V) for the driver circuit when load gates are not connected . Then also find out the power dissipation (of the driver) for this case.	2
(b)	Find I_1 , I_2 , I_{B1} , I_{C1} and I_D if any input is logical high (10 V) for the driver circuit when load gates are not connected . Then also find out the power dissipation (of the driver) for this case.	2
(c)	The driver circuit is now connected with Load gates. Calculate maximum fanout of the overall DTL setup.	1

Question 3

For the transistors in the DTL & RTL circuit in figure 3 and 4 assume $\beta_F = 65$, $V_{BE}(\text{saturation}) = 0.8V$, and $V_{CE}(\text{saturation}) = 0.2V$. The input voltages A and B can be 0V or 5V.

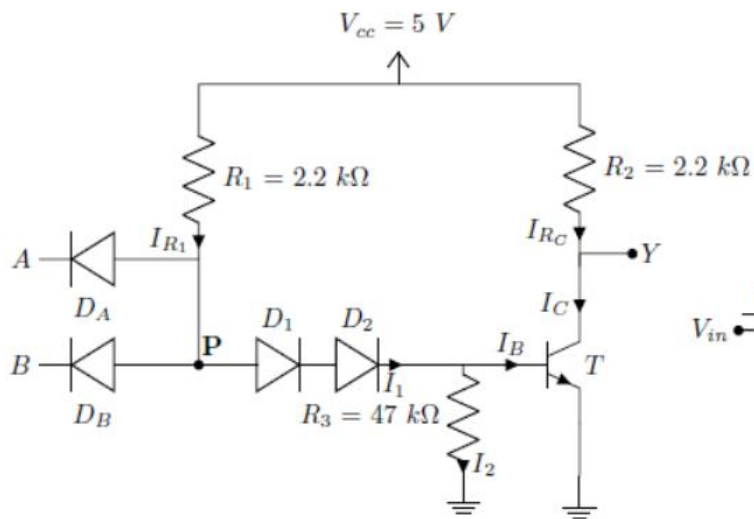


Figure 3: DTL NAND gate

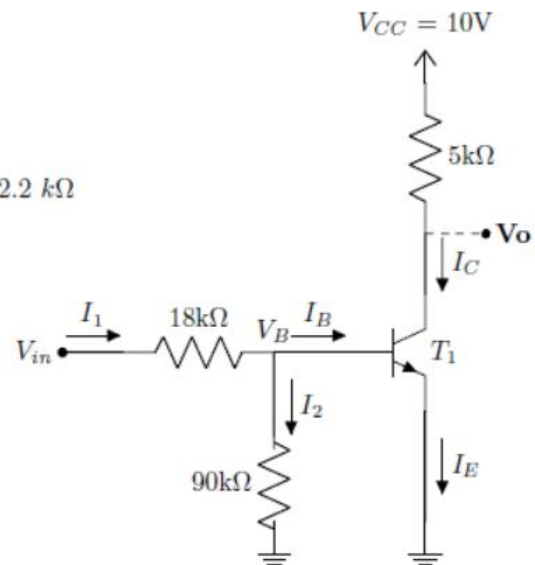


Figure 4: RTL for part (c) only

Only consider figure 3 for part (a) & part (b).

(a)	Calculate the maximum fanout when the DTL circuit is connected to identical loads, the same as the driver circuit in figure 3 and the inputs are set to $A = B = 5V$.	2
(b)	Calculate the total power dissipation of the DTL circuit in figure 3 when $A = 5V$ & $B = 0V$.	1

The designer now wants to make an AND gate by using the DTL-NAND gate (figure 3) as the driver and the RTL-NOT gate (figure 4) as the load.

(c)	Calculate the maximum number of RTL NOT gates a single DTL NAND gate can drive when the input of the DTL is both high ($A = 5V$, $B = 5V$).	2
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Question 4

For the RTL circuit in Figure 5, assume $V_{BE}(\text{saturation}) = 0.8\text{V}$, $V_{CE}(\text{saturation}) = 0.2\text{V}$, $V_{\gamma} = 0.5\text{V}$, $\beta_F = 30$, $V_{OH} = 10\text{V}$, $V_{OL} = 0.2\text{V}$

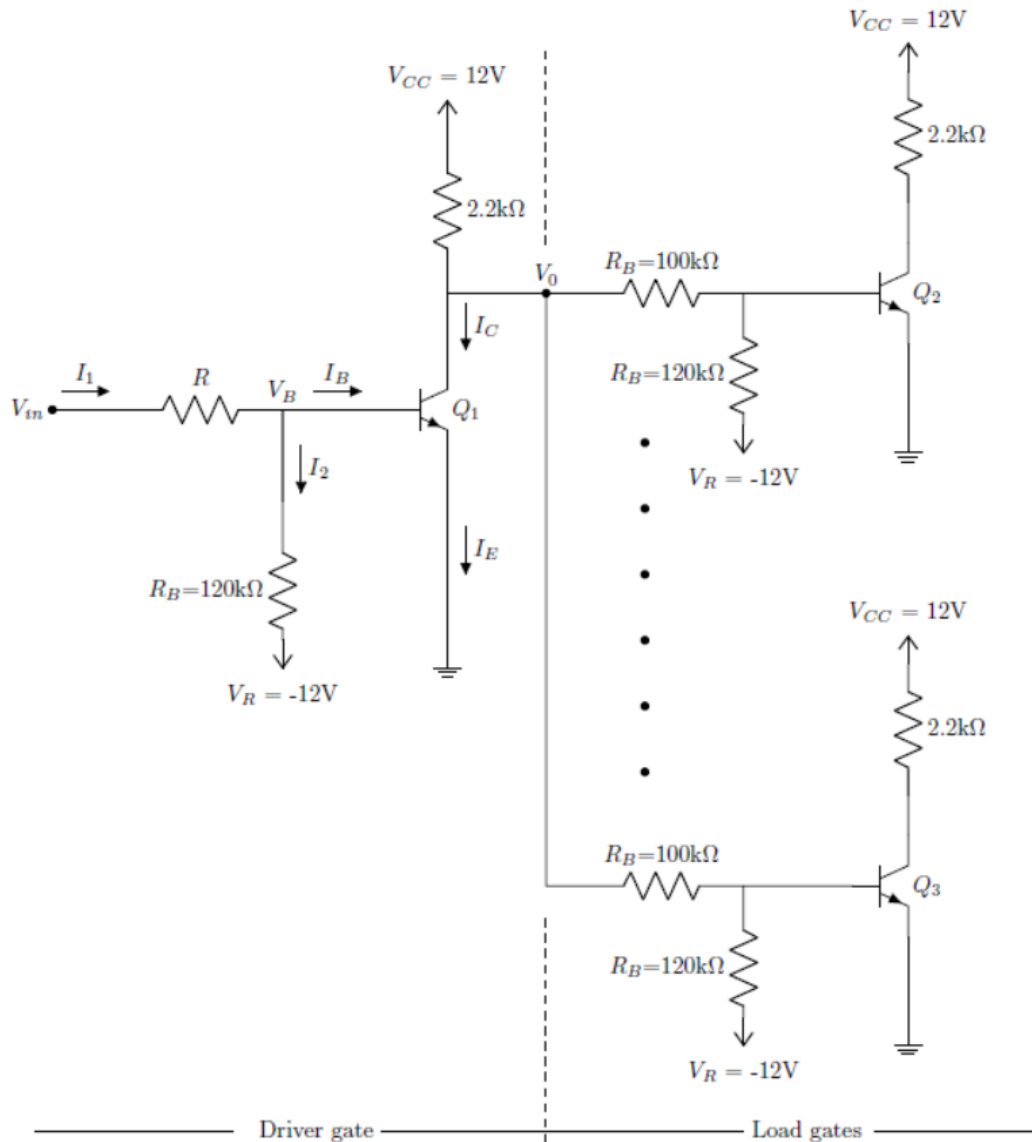


Figure 5

(a)	Consider no loads are attached to the driver circuit, determine the total power dissipated in the driver when input, $V_{in} = 0.2\text{V}$ [Assume $R = 10\text{k}\Omega$].	2
(b)	For high noise margin (V_{NH}) of 6V, estimate the value of R in $\text{k}\Omega$ when no loads are connected to the driver.	2
(c)	Determine the maximum possible fanout [Assume $R = 10\text{k}\Omega$].	1